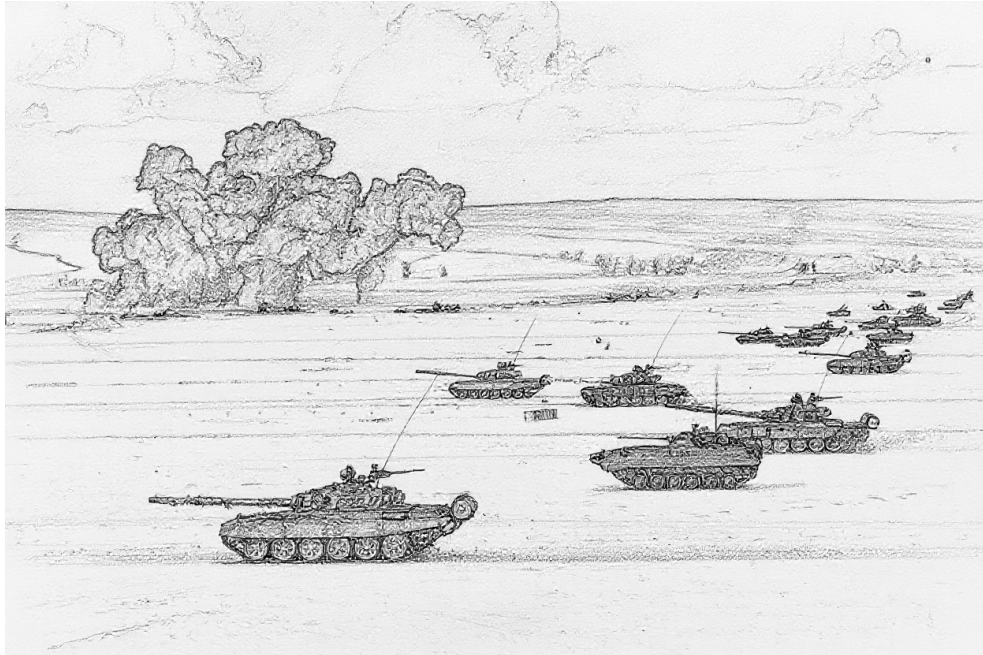


Air Power:

Ground Unit and Ground Target Data

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1 May 2026



INTRODUCTION

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:¹

“You’re not going to win a war by shooting down a damn MiG, you’re going to win a war by bombing the bejesus out of whatever you were sent to bomb.”

Here we deal with those “whatevers,” the ground units and ground targets that are the targets of air-to-ground attacks, and some of which shoot back.

This document first presents ground-combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from *Air Power Journal*, and from Malcolm Pipe’s extensive compendium of AAA and SAM units. However, the data have been carefully revised to improve their accuracy and internal consistency, a wider variety of ground-combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded.

Extensive notes accompany the main text, providing provenance of data, explanations of changes, sources of information, and justifications for decisions.

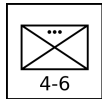
GROUND-COMBAT AND TRANSPORT UNITS

Ground-combat units are formations of infantry, tanks, IFVs, APCs, other AFVs, or artillery whose primary role is combat with other ground units. Transport units provide strategic mobility to infantry and towed artillery and are the life-blood of logistic networks. These units are presented below with their counters. The Ground-Combat and Transport Unit Table summarizes their properties. All ground-combat and transport platoons and batteries are also available as sections and squads.²

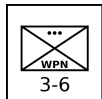
INFANTRY UNITS

Infantry are soldiers trained to fight on foot and, in some cases, from IFVs. Their primary roles in modern warfare are seizing, clearing, and holding ground, often as part of a combined-arms formation with tanks and artillery, highly mobile offensive and defensive operations in combination with helicopters, and combat in difficult terrain, including urban areas, mountainous zones, and dense jungle or forest.

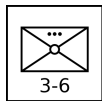
An important secondary characteristic of infantry is any associated transport. Motorized infantry units are transported by trucks, giving them operational mobility. Mechanized infantry are transported by APCs or IFVs, giving them both tactical and operational mobility. Light infantry units have no or fewer supporting vehicles, although modern light infantry is often airborne.



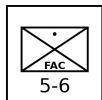
Infantry Platoon: An infantry platoon is armed principally with small arms, often supplemented with light machine guns and sometimes with light grenade launchers and portable anti-tank weapons such as bazookas, LAWs, MAWs, and portable ATGMs. It is the primary combat unit in larger infantry formations.



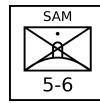
Infantry Weapons Platoon: An infantry weapons platoon is equipped with heavier weapons, such as medium or heavy machine guns, recoilless rifles, portable ATGMs, and automatic grenade launchers. It is often a company- or battalion-level asset, and its role is to provide direct-fire support to the infantry platoons in its formation.³



Infantry Mortar Platoon: An infantry mortar platoon is equipped with 60-mm, 81-mm, 82-mm, or similar mortars. It is often a battalion-level asset, and its role is to provide indirect fire support and smoke-screens to the infantry platoons in its formation. Due to the weight of ammunition required for sustained fire, it will often be transported by a light truck platoon even if the other infantry units in its formation do not have transport.

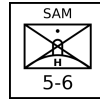


Infantry FAC Squad: An infantry FAC squad has a forward air controller with other soldiers providing communication and protection. An infantry FAC squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



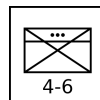
Infantry SAM Squad: Infantry SAM squads are equipped with shoulder-launched SAMs, with one launcher per squad. Their SAM capabilities and VP values are given elsewhere.

An infantry SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry Heavy SAM Squad: Infantry heavy SAM squads are equipped with pedestal-mounted SAMs, with one launcher per squad. Their SAM capabilities and VP values are given in a subsequent section.

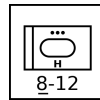
An infantry heavy SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry HQ Platoon: An infantry HQ platoon is the HQ unit of an infantry battalion or regiment. It consists of the formation commander, their staff, and their protection detail.

ARMORED VEHICLE UNITS

Armored ground-combat vehicle units combine armored protection for their crew and weapons, mobility, and either a fire-power or transport capability. The units described here are generic, but the Vehicle Types Table gives the correspondence between specific vehicles and these generic vehicle types. Armored cars and armored half-tracks are considered to be light tanks, light anti-tank missile vehicles, light reconnaissance vehicles, light IFVs, or light or open APCs according to their role and armament. That is, the means of traction — tracked, half-tracked, or wheeled — is a secondary consideration.



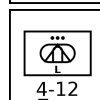
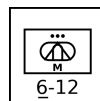
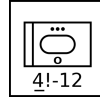
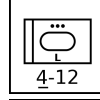
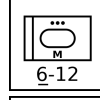
Tank Platoon: A tank platoon has armored vehicles principally armed with a direct-fire gun. The gun is most commonly mounted in a turret, but AFVs with casemate guns are also considered to be tanks. Its role is the support of infantry and the destruction of other tanks and armored vehicles. A tank platoon may have heavy, medium, light, or open armor.⁴

Heavy examples: M1 Abrams and T-64/72/80.

Medium examples: M60 and T-54/55/62.

Light examples: M551 Sheridan and PT-76.

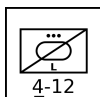
Open example: ASU-57.



Anti-Tank Missile Platoon: An anti-tank missile platoon has armored vehicles equipped with anti-tank guided missiles. Its mission is the destruction of other armored vehicles. Conventional gun-armed tank destroyers are considered to be tanks. It is often a battalion- or regiment-level asset. An anti-tank platoon may have medium or light armor.⁵

Medium example: Rakaten Jpz.

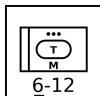
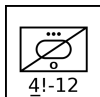
Light examples: M901 and BRDM.



Reconnaissance Platoon: A reconnaissance platoon has armored vehicles specialized in scouting and observation and have no offensive weapons more powerful than machine guns. Vehicles used for reconnaissance that do have significant offensive weapons (e.g. 20 mm or larger guns) are typically classified as light tanks. A reconnaissance platoon may have light or open armor.⁶

Light examples: M114 and BRDM-2.

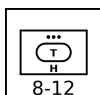
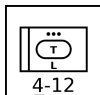
Open example: M20.



IFV Platoon: An IFV platoon is equipped with armored vehicles that transport infantry and provide fire support. They are distinguished from APCs by their armament of a 20 mm or larger cannon and sometimes an anti-tank missile. Some IFVs are also designed to allow their infantry to fight while mounted. An IFV platoon may have medium or light armor.⁷

Medium example: M2A2 Bradley.

Light examples: M2/M2A1 Bradley and BMP-1/2.



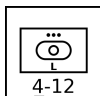
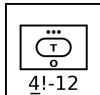
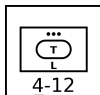
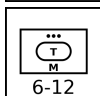
APC Platoon: An APC platoon is equipped with armored vehicles that transport infantry, but unlike IFVs are not equipped to provide fire support and at most are armed with a heavy machine gun. An APC platoon may have heavy, medium, light, or open armor.⁸

Heavy examples: Nakpadon.

Medium examples: Nagmashot and Nagmachon.

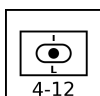
Light examples: M113 and BTR-50PK/60PB/70.

Open examples: M3 half-track and BTR-50P/60P.



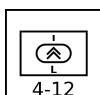
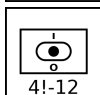
Armored Mortar Platoon: An armored mortar platoon has mortars mounted in armored vehicles (typically converted APCs or IFVs). It is typically a battalion-level in mechanized-infantry or tank battalions, and its role is indirect fire support. An armored mortar platoon has light armor.⁹

Examples: M106 and M125.



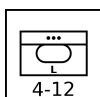
Armored Artillery Battery: An armored artillery battery has guns or howitzers in armored vehicles. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support. An armored artillery batteries may have light or open armor.¹⁰

Examples: M109, 2S3, and 2S1.



Armored Rocket Artillery Battery: An armored rocket artillery battery is the rocket-armed equivalent of an armored artillery battery. An armored rocket artillery battery has light armor.¹¹

Example: M270.



Armored HQ Platoon: An armored HQ platoon is the HQ unit of a tank, mechanized-infantry, or armored-artillery battalion, regiment, or brigade. It consists of the formation commander, their staff, and their protection detail. It is typically equipped with modified APCs or IFVs. An armored HQ platoon has light armor.

UNARMORED VEHICLE UNITS

Unarmored or soft vehicles provide mobility, but do not provide armor protection (or provide only incomplete armor protection) for their crew and weapons. The units described here are generic, but the Vehicle Types Table gives the correspondence between specific vehicles and these generic vehicle types.



Truck Platoons: A truck platoon has unarmored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition.



Light Truck Platoons: A light truck platoon has unarmored wheeled vehicles that can transport light loads of infantry, towed weapons, and supplies, including fuel and ammunition.

Examples: M151 Jeep and GAZ-69.



Tracked Carrier Platoons: A tracked carrier platoon is equipped with unarmored tracked vehicles that can transport medium loads of infantry, towed weapons, and supplies, including fuel and ammunition.¹²

Example: Bv 202.



Tracked Artillery Battery: A tracked artillery battery has guns or howitzers mounted on tracked vehicles that provide mobility, but do not give at most limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support.¹³

Examples: M107, M110, 2S4, and 2S7.



Mobile Artillery Battery: A mobile artillery battery has howitzers mounted on wheeled vehicles that provide mobility, but give at most limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support.¹⁴

Examples: CAESAR and Archer.



Mobile Rocket Artillery Battery: A mobile rocket artillery battery is the rocket-armed equivalent of a mobile artillery battery.¹⁵

Examples: M142 HIMARS and BM-21 Grad.



Tracked Missile Artillery Battery: A tracked rocket artillery battery is the rocket-armed equivalent of a mobile tracked battery.¹⁶

Example: M752 Lance.



Mobile Missile Artillery Battery: A mobile missile artillery battery is the missile-armed equivalent of a mobile artillery battery.¹⁷

Examples: M1003 Pershing II and OTR-21 Tocha (SS-21).

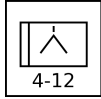


Mobile HQ Platoon: A mobile HQ platoon is typically the HQ unit of a battalion, regiment, or brigade other than those of infantry, tank, mechanized infantry, or armored artillery. It is equipped with trucks, and its personnel are not specialized in fighting as infantry.¹⁸

TOWED UNITS

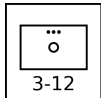
Towed weapons provides neither inherent mobility nor protection. However, they are light and cheap compared to simi-

lar vehicle-mounted weapons. Towed weapons is typically used away from direct combat and provided with appropriate ground or helicopter transport.



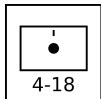
Towed Anti-Tank Gun Battery: A towed anti-tank gun battery has anti-tank guns on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.

Example: 2A18 (T-12).



Towed Mortar Platoon: A towed mortar platoon has 120 to 160 mm mortars on unprotected, towed mounts. It is often a regimental- or brigade- asset. Smaller infantry mortars are found in infantry mortar platoons.

Example: M1943.



Towed Artillery Battery: A towed artillery battery has guns or howitzers on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.

Examples: M119, M198, and 2A18 (D-30).

Ground-Combat and Transport Unit Table

Type	Size ^a	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class ^b
Infantry	Platoon	4	6	U	5/3/2	B2
Infantry Weapons	Platoon	3	6	U	6/4/2	B2
Infantry Mortar	Platoon	3	6	U	6/4/2	B2
Infantry FAC	Squad	5	6	U	-/-/8	—
Infantry SAM	Squad	5	6	U	—	—
Infantry Heavy SAM	Squad	5	6	U	—	—
Infantry HQ	Platoon	4	6	U	10/7/3	B2
Heavy Tank	Platoon	8	12	U	12/8/4	B3
Medium Tank	Platoon	6	12	U	10/7/3	B3
Light Tank	Platoon	4	12	U	6/4/2	B3
Open Tank	Platoon	4	12	U	5/3/2	B3
Medium Anti-Tank Missile	Platoon	6	12	U	12/8/4	B3
Light Anti-Tank Missile	Platoon	4	12	U	10/7/3	B3
Light Reconnaissance	Platoon	4	12	U	6/4/2	B3
Open Reconnaissance	Platoon	4	12	U	4/3/1	B3
Medium IFV	Platoon	6	12	U	10/7/3	B3
Light IFV	Platoon	4	12	U	6/4/2	B3
Heavy APC	Platoon	8	12	U	10/7/3	B3
Medium APC	Platoon	6	12	U	8/5/3	B3
Light APC	Platoon	4	12	U	6/4/2	B3
Open APC	Platoon	4	12	U	5/3/2	B3
Light Armored Mortar	Platoon	4	12	U	10/7/3	B3
Light Armored Artillery	Battery	4	12	U	12/8/4	B3
Open Armored Artillery	Battery	4	12	U	12/8/4	B3
Light Armored Rocket Artillery	Battery	4	12	U	12/8/4	B3
Light Armored HQ	Platoon	4	12	U	12/8/4	B3
Truck	Platoon	2	12	G/P	4/3/1	—
Light Truck	Platoon	2	12	G/P	4/3/1	—
Tracked Carrier	Platoon	2	12	U	4/3/1	—
Mobile Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Artillery	Battery	2	12	U	10/7/3	—
Mobile Rocket Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Missile Artillery	Battery	2	12	U	10/7/3	—
Mobile Missile Artillery	Battery	2	12	G/P	9/6/3	—
Mobile HQ	Platoon	2	12	G/P	10/7/3	—
Towed Mortar	Platoon	3	12	T	7/5/2	—
Towed Anti-Tank Gun	Battery	4	12	T	8/5/3	—
Towed Artillery	Battery	4	18	T	8/5/3	—

^a Platoons and batteries are available as sections and squads with corresponding modifications to their damage resiliences and VPs.

^b Platoons with AAA class of B2 and B3 can employ barrage fire to altitudes of 2 and 3 levels, respectively.

Vehicle Types Table

Heavy Tanks		Light Tanks		Medium IFVs		Heavy APCs		Light Armored Mortar Carriers	
Al-Khalid/VT-1A	PK/CN 2001	AMX-10 RC	FR 1981	BMP-3	SU 1987	Achzarit	IL 1988	FV432 with 81 mm	GB 1963
Ariete	IT 1995	AMX-13	FR 1953	Boxer IFV	NL/DE 2011	Nakpadon	IL 1998	M106	US 1964
Arjun	IN 2004	Fox	GB 1975	Strf 9040/CV90	SE 1994	Namer	IL 2008	M125	US 1964
Challenger 1	GB 1983	Commando (90 mm)	US 1964	LAV III (add-on-armor)	CA 2009	Puma	IL 1993	M1064	US 1990
Challenger 2	GB 1998	LAV I Cougar	CA 1976	M2A2/3/4 Bradley	US 1988			Stryker MCV	US/CA 2005
Chieftain	GB 1967	LAV II Coyote	CA 1990	M3A2/3/4 Bradley	US 1988				
Conqueror	GB 1955	M8/Greyhound	US 1943	SPz Marder	DE 1971				
IS-2	SU 1944	M24 Chaffee	US 1944	SPz Puma	DE 2015				
IS-3	SU 1945	M41 Walker Bulldog	US 1953						
K1	KR 1987	M113 C&V with 25 mm	US 1974						
K2 Black Panther	KR 2014	M114A2	US 1969						
Leclerc	FR 1992	M551 Sheridan	US 1969						
Leopard 2	DE 1979	Panhard AML	FR 1961						
M-84	YU 1985	Panhard EBR	FR 1951						
M1 Abrams	US 1980	PT-76	SU 1951						
Merkava 1	IL 1979	Saladin	GB 1958						
Merkava 2	IL 1983	Scimitar	GB 1974						
Merkava 3	IL 1989	Scorpion	GB 1973						
Merkava 4	IL 2004	SpPz Luchs	DE 1975						
PT-91	PL 1995	Stryker MGS	US/CA 2006						
Stridsvagn 103	SE 1967	Strv 74	SE 1958						
T-10	SU 1954	Type 62	CN 1961						
T-14 Armata	RU 2024	Type 63	CN 1963						
T-64	SU 1966	Type 05 AAV	CN 2005						
T-72	SU 1973	Type 08 ARV	CN 2008						
T-80	SU 1976								
T-84	UA 1999								
T-90	RU 1992								
Type 10	JP 2012								
Type 90	JP 1990								
Type 99	CN 2001								
Medium Tanks									
AMX-30	FR 1966	Raketen JPz 1	DE 1961						
ASU-85	SU 1959	Raketen JPz 2	DE 1967						
Centurion	GB 1945	Jaguar 1	DE 1978						
Comet	GB 1944	Jaguar 2	DE 1983						
Cromwell	GB 1944								
ISU-122	SU 1944	BRDM-2 ATGM	SU 1973						
ISU-152	SU 1943	BRDM-3 ATGM	SU 1973						
Kanonen JPz	DE 1965	Fennek MRAT	DE/NL 2003						
Leopard 1	DE 1965	FV438 Swingfire	GB 1971						
M-50 Sherman	IL/FR 1956	M150	US 1973						
M-51 Sherman	IL/FR 1961	M901 ITV	US 1979						
M103	US 1957	Striker	GB 1976						
M26 Pershing	US 1944	Stryker ATGM	US/CA 2003						
M4 Sherman	US 1942	VAB HOT	FR 1978						
M46 Patton	US 1949	YP-408 PRAT	NL 1976						
M47 Patton	US 1951	YPR-765 PRAT	NL 1980						
M48 Patton	US 1952								
M60	US 1959								
Panzer 61	CH 1965								
Panzer 68	CH 1971								
SU-100	SU 1944								
T-34	SU 1940								
T-54	SU 1955								
T-55	SU 1958								
T-62	SU 1961								
TAM	AR 1983								
Type 59	CN 1959								
Type 61	JP 1961								
Type 69	CN 1972								
Type 74	JP 1975								
Type 79	CN 1984								
Type 85	CN 1988								
Type 88	CN 1988								
Type 96	CN 1997								
Type 155	CN 2018								
Vickers MBT	GB 1965								
Vijayanta	IN/GB 1965								

EXAMPLE ORDERS OF BATTLE

This appendix contains a number of example orders of battle and their correspondence in game units.

– Table 1: 2 Para Battalion at the 1982 Battle of Goose Green. At this time, the battalion had attached half a battery of 105 mm Light Guns from Alma Battery of 29 Commando Regiment, RA, a section of Blowpipes from 43rd Air Defence Battery, RA, and a Recce Troop of 59 Independent Commando Squadron, RE.

– Table 2: 12th Infantry Regiment at the 1982 Battle of Goose Green, with attachments from the 25th Infantry Regiment, 4th Airborne Artillery Regiment, 601st Anti-Aircraft Artillery Group.

– Tables 3, 4, 3, 5, 6, and 7: An armored regiment, APC infantry battalion, infantry battalion, field artillery regiment, and heavy artillery regiment of the British Army from about 1960. In the British Army, regiments are battalion-level units.¹⁹

– Table 8: An armored regiment of the British Army from about 1985.

– Tables 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, and 22: A Soviet front-line motor-rifle division from the 1980s with its constituent regiments and battalions.²⁰ In the Soviet Army, regiments are brigade-level units.

Table 1: 2 Para at Goose Green

Element	Game Ground Unit	Notes
Battalion HQ	1 × infantry HQ platoon	
A Company	3 × infantry platoon	
B Company	3 × infantry platoon	
C Company	2 × infantry platoon	Patrol Company
D Company	3 × infantry platoon	
Support Company	1 × infantry mortar platoon	
	1 × infantry weapons platoon	GPMGs
	1 × infantry weapons platoon	MILANs
	1 × infantry platoon	Engineers
Alma Battery	1 × towed artillery section	Three 105-mm Light Guns
Blowpipe Section	2 × infantry SAM squad – Blowpipe	
Engineers Troop	1 × infantry platoon	

Table 2: 12IR at Goose Green

Element	Game Ground Unit	Notes
Regimental HQ	1 × infantry HQ platoon	
SAM Section	2 × infantry SAM squad – SA-7B	
A Company 12IR	5 × infantry platoon	
B Company 12IR	3 × infantry platoon	
C Company 25IR	2 × infantry platoon	
	1 × infantry weapons platoon	
C Company 12IR	2 × infantry platoon	
	1 × infantry weapons platoon	
A Battery, 4AAR	1 × towed artillery section	Three 105-mm Pack Howitzers
B Battery, 601 GADA	1 × Oerlikon 35 mm GDF section	Two guns
	1 × towed FCR-D	Skyguard
9th Engineering Company	1 × infantry platoon	
3 Section B Battery 601AAAG	1 × Rh-202 20 mm battery	Six guns
	1 × EWR	Elta
Security Company	1 × infantry platoon	

Table 3: British Army Armored Regiment (Early 1960s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
1 × Reconnaissance Squadron	3 × light reconnaissance/tank platoon	15 × Ferret/Saladin
3 × Tank Squadrons (A/B/C) each:	3 × medium tank platoon	14 × Centurion
	1 × heavy tank platoon	3 × Conqueror

Table 4: British Army APC Infantry Battalion (Early 1960s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
3 × Infantry Companies (A/B/C) each:	3 × infantry platoon	
	1 × infantry weapons platoon	
	4 × light APC platoon	17 × Saracen/FV432

Table 5: British Army Infantry Battalion (Early 1960s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × infantry HQ platoon	
	1 × truck platoon	
4 × Infantry Companies (A/B/C/D) each:	3 × infantry platoon	
	1 × infantry weapons platoon	
	4 × truck platoon	

Table 6: British Army Field Artillery Regiment (Early 1960s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × mobile HQ platoon	
4 × Batteries (A/B/C/D) each:	1 × tracked artillery battery	6 × Cardinal (M44)

Table 7: British Army Heavy Artillery Regiment (Early 1960s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × mobile HQ platoon	
2 × Batteries (A/B) each:	1 × mobile rocket section	2 × Honest John
2 × Batteries (C/D) each:	1 × towed artillery battery	4 × 8-inch Gun M1
	1 × truck platoon	

Table 8: British Army Armored Regiment (Mid 1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
1 × Reconnaissance Troop	2 × light tank platoon	8 × Scorpion
1 × ATGW Troop	3 × light anti-tank platoon	9 × FV438
3 × Tank Squadrons (A/B/C) each:	4 × heavy tank platoon	14 × Chieftain

Table 9: Soviet Tank Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
3 × Tank Companies each:	3 × heavy tank platoon	12 × T-64/72/80

Table 10: Soviet BMP Motor Rifle Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
1 × Mortar Battery	1 × towed mortar platoon	M1943 120 mm mortar
	1 × truck platoon	
1 × Grenade Launcher Platoon	1 × light IFV/APC platoon	3 × BMP-1/2
	1 × infantry weapons platoon	6 × AGS-17 grenade launchers
3 × Motor Rifle Companies each:	4 × light IFV platoon	12 × BMP-1/2
	3 × infantry platoon	
	1 × infantry weapons platoon	6 × PKM machine guns
	3 × SA-7/14/16 squad	3 × SA-7/14/16

Table 11: Soviet BTR Motor Rifle Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
1 × Mortar Battery	2 × towed mortar platoon	8 × M1943 mortar
	2 × truck platoon	
1 × Anti-Tank Platoon	1 × light IFV/APC platoon	5 × BTR-60/70
	1 × infantry weapons platoon	2 × recoilless rifle and 4 × ATGM
1 × Grenade Launcher Platoon	1 × light IFV/APC platoon	3 × BTR-60/70
	1 × infantry weapons platoon	6 × AGS-17 grenade launchers
3 × Motor Rifle Companies each:	4 × light APC platoon	12 × BTR-60/70
	3 × infantry platoon	
	1 × infantry weapons platoon	3 × PKM machine gun and 3 × ATGM
	3 × SA-7/14/16 squad	3 × SA-7/14/16

Table 12: Soviet Self-Propelled Artillery Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × mobile HQ platoon	
3 × Howitzer Battery each	1 × towed artillery battery	6 × 122-mm 2S1 Gvozdika

Table 13: Soviet Artillery Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × mobile HQ platoon	
3 × Howitzer Battery each	1 × towed artillery battery	6 × 122-mm D-30
	1 × truck platoon	

Table 14: Soviet Tank Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon 3 × SA-7/14/16 squad 3 × light truck squad	
1 × Reconnaissance Company	1 × light IFV platoon 1 × light reconnaissance platoon	3 × BMP-1/2 4 × BRDM-2
1 × Air-Defense Battery	2 × SA-9/13 section 2 × ZSU-23-4 section	4 × SA-9/13 4 × ZSU-23-4
3 × Tank Battalion		
1 × Self-Propelled Artillery Battalion		

Table 15: Soviet BMP Motor-Rifle Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon 3 × SA-7/14/16 squad 3 × light truck squad	
1 × Reconnaissance Company	1 × light IFV platoon 1 × light reconnaissance platoon	3 × BMP-1/2 4 × BRDM-2
1 × Anti-Tank Battery	3 × light anti-tank platoon	9 × BRDM-2 ATGM
1 × Air-Defense Battery	2 × SA-9/13 section 2 × ZSU-23-4 section	4 × SA-9/13 4 × ZSU-23-4
3 × BMP Motor-Rifle Battalions		
1 × Tank Battalion		
1 × Self-Propelled Artillery Battalion		

Table 16: Soviet BTR Motor-Rifle Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon 3 × SA-7/14/16 squad 3 × light truck squad	
1 × Reconnaissance Company	1 × light IFV platoon 1 × light reconnaissance platoon	3 × BMP-1/2 4 × BRDM-2
1 × Anti-Tank Battery	3 × light anti-tank platoon	9 × BRDM-2 ATGM
1 × Air-Defense Battery	2 × SA-9/13 section 2 × ZSU-23-4 section	4 × SA-9/13 4 × ZSU-23-4
3 × BTR Motor-Rifle Battalions		
1 × Tank Battalion		
1 × Artillery Battalion		

Table 17: Soviet Artillery Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon	
3 × Self-Propelled Artillery Battalions each	3 × light armored artillery battery	18 × 152-mm 2S3
1 × Rocket Artillery Battalion	3 × mobile rocket artillery battery	18 × BM-21

Table 18: Soviet SAM Regiment (1980s)

Element	Game Ground Unit	Notes
1 × Regimental HQ	1 × armored HQ platoon 3 × SA-7/14/16 squad 3 × light truck squad	
1 × EWR Battery	2 × EWR-B	2 × Long Track EWR and 1 × Thin Skin HFR
5 × SAM Batteries each	1 × SA-6 battery 1 × SA-7/14/16 squad 1 × light truck squad	4 × SA-6 TELAR and 1 × Straight Flush TTR

Table 19: Soviet Reconnaissance Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
2 × Reconnaissance Company each	2 × light IFV platoon 2 × infantry platoon 1 × heavy tank platoon	6 × BMP-1/2 3 × T-64/72/80
1 × Reconnaissance Assault Company	3 × light reconnaissance platoon	18 × BRDM-2

Table 20: Soviet Anti-Tank Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × armored HQ platoon	
2 × Anti-Tank Gun Battery each	1 × towed anti-tank battery 1 × light APC platoon	6 × 100-mm MT-12 7 × MT-LB
1 × ATGM Battery	3 × light anti-tank platoon	9 × BRDM-2 ATGM

Table 21: Soviet SSM Battalion (1980s)

Element	Game Ground Unit	Notes
1 × Battalion HQ	1 × mobile HQ platoon	
2 × SSM Battery each	1 × mobile missile artillery section	2 × FROG-7 or SS-21

RULE 1: GROUND UNITS

A ground *unit* is a combat or logistics formation of infantry, armor, transports, AAA, SAM launchers, or radars. This section discusses them in general.²¹

A. Type and Size: A ground unit will have a type, for example, “heavy tank” or “ZSU-23-4”, and a size, which will be one of squad, section, platoon, or battery. Typically, a squad has five to ten soldiers and/or one vehicle, gun, or launcher; a section has ten to twenty soldiers and/or two vehicles, guns, or launchers; and a platoon or battery has thirty to fifty soldiers and/or three to six vehicles, guns, or launchers. The term “platoon” is more commonly used for infantry and tank units and “battery” for artillery and air-defense units, but within the game they are equivalent.

Companies, battalions, and other larger units above platoons or batteries are represented by multiple units. For example, an infantry company might be represented by three infantry platoon units.

All ground-combat and transport platoons are also available as sections and squads and all AAA platoons and batteries are also available as sections. A section or squad is identical in every way to a platoon or battery that has taken 1D or 2D damage, respectively, except that no VPs are awarded for this notional damage and non-AAA squads and sections may not engage in barrage fire.²²

In the game, ground units are uniformly referred to as squads, sections, platoons, and batteries. However, other terms are used for these units even within English-speaking armies. For example, in many Commonwealth armies a section is equivalent to a squad, a troop is equivalent to a platoon, and a squadron is equivalent to a company. In the US Armored Cavalry, a troop is equivalent to a company and a squadron is equivalent to a battalion.

B. Basic Properties: The type and size of a ground unit determines its basic properties: damage resilience, defense strength, target class, mobility, and sighting range.

The damage resilience is the amount of damage a ground unit can suffer before being killed. It is determined by the size of the unit: a squad has a damage resilience of 1D; a section has a damage resilience of 2D; and a platoon or battery has a damage resilience of 3D.²³

The numerical defense strength is used to calculate the odds for air-to-ground attacks and for ground-to-ground combat.

The target class is used to determine the appropriate attack strength of any weapons used in air-to-ground attacks. The class can be a hard, open, or soft according to the degree to which the unit enjoys armored protection. Many weapons have different attack strengths against targets of different classes. A hard target has armored protection from both direct (front, side, and rear) and top attacks. Hard targets are further described as light, medium, or heavy according to their degree of protection, and have corresponding defense strengths.²⁴ An open target has armored protection from direct attacks but not from top attacks. An open target is normally treated as a hard target, but against napalm, fire, FAE, and air-burst HE bombs, it is considered to be a soft target.²⁵ A soft target does not have armored protection their personnel and/or weapons.²⁶

C. Representation: The briefing documents show the graphical representation of ground units as counters.²⁷ The size of a unit is indicated by one, two, or three dots (for squads, sections, or platoons) or one vertical bar (for batteries) in an upper position.

Hard targets are indicated by an underlined defense strength and open targets by an underline defense strength followed by an exclamation point.

Advanced Rule

D. Mobility: The mobility of a ground unit restricts the hexes it can occupy according to the terrain in and around those hexes. Infantry units, armored units (whether tracked or wheeled), and tracked unarmored units have unrestricted (U) mobility. Trucks and other wheeled unarmored units have good (G) or poor (P) mobility. All other units have towed (T) mobility.²⁸

A unit with unrestricted mobility may occupy: any urban hex; any clear hex; or any forest hex. A unit with good mobility may occupy: any urban hex; any clear hex; or any forest hex with a road or trail. A unit with poor mobility may occupy: any urban hex; any clear or forest hex with a road, trail, or runway; or any clear hex adjacent to an urban hex or another clear hex with a road, trail, or runway. A unit with towed mobility may occupy hexes according to the mobility of its transport.²⁹

When necessary, a scenario must specify whether wheeled unarmored units have good or poor mobility, according to the terrain and the capabilities of the vehicles represented, and assign transport units to towed units.³⁰ A scenario may modify the mobility classes of ground units and the associated restrictions.³¹

E. Composite Platoons: At the start of the game, an infantry, infantry weapons, or infantry HQ platoon may attach one or two infantry squads to form a composite platoon. The attached squads are commonly infantry SAM or FAC squads. Once formed, a composite platoon may not be split into its constituent units.³²

An attached squad is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. Instead, it is considered to form part of the composite platoon. An attached squad retains its normal capabilities (for example, an attached SAM squad may engage aircraft and an attached FAC squad may sight and mark targets).

A composite platoon has the same damage resilience, defense strength, target class, mobility, and sighting range as its constituent platoon. If a composite platoon is sighted, no information is given on any attached squads. If a composite platoon is identified, any attached squads are also identified (for example, “infantry platoon with an attached FAC squad”). If a composite platoon is suppressed, each attached squad is suppressed too. If a composite platoon suffers damage, then for each D of damage, roll one die for each attached squad and on a 3– the attached squad is killed. If a composite platoon is killed, each attached squad is killed too. VPs are awarded for damage to each of the constituent units.³³

F. Transport: An infantry or towed unit may be transported by a specific associated transporting IFV, APC, truck, or tracked-carrier unit of at least the same size. When necessary, a scenario will establish the association between units and their specific transports.³⁴

A transported unit is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. It may not attack other ground units or aircraft, may not launch SAMs, may not use radar, and may not act as a FAC.

If a transporting unit is sighted, no information is given on the transported unit. If a transporting unit is identified, no information is given on a transported infantry unit, but any transported

towed unit is also identified (for example, “truck platoon towing an artillery battery” or “truck platoon towing a ZPU-4 battery”). If a transporting unit suffers damage, then its transported unit suffers the same damage. VPs are awarded for damage to both the transporting and transported unit.

A unit may start the game either transported or not. If not transported, it may be placed according to either its mobility or that of its associated transport.³⁵

An infantry unit in the same hex as its associated transport may declare that it is mounting the transport at the start of a Ground Unit Interaction Phase. Both units must not be suppressed, and the transport must not have suffered more damage than the infantry unit. At the end of the Ground Unit Interaction Phase five game turns later, the unit is considered to be transported, and its counter is removed from the map.

An infantry unit being transported may declare that it is dismounting at the start of a Ground Unit Interaction Phase. The transport unit must not be suppressed. At the end of the Ground Unit Interaction Phase two game turns later, the unit is considered to be no longer transported, and its counter is placed on the map in the same hex as its transport. If placing the counter would violate stacking requirements, the unit must abort the dismounting process.

During these processes, the infantry unit may not perform any other action and the transport may not move and may not carry out ground attacks, but may perform barrage fire if it has this capability. The infantry unit may abort either of these process at the start of any Ground Unit Interaction Phase. If either are suppressed during the process, the unit must abort the process in the next Ground Unit Interaction Phase.³⁶

A towed unit may not mount or dismount during the game.

Table 22: Soviet Motor-Rifle Division (1980s)

Element	Game Ground Unit	Notes
1 × Divisional HQ	1 × armored HQ platoon 6 × SA-7/14/16 squad 6 × light truck squad	
1 × Reconnaissance Battalion		
1 × Anti-Tank Battalion		
1 × SSM Battalion		
1 × BMP Motor-Rifle Regiment		
2 × BTR Motor-Rifle Regiment		
1 × Tank Regiment		
1 × Artillery Regiment		
1 × SAM Regiment		

NOTES

1. Robin Olds.

The full quotation contains a second insight: “You’re not going to win a war by shooting down a damn MiG, you’re going to win a war by bombing the bejesus out of whatever you were sent to bomb. Of course, we weren’t there to win it anyway, so it was all a waste of time.” (Sherwood, John. *Fast Movers: Jet Pilots and the Vietnam Experience*, chapter 1)

2. Ground-Combat and Transport Units.

Air Strike and *Eagles of the Gulf* provide eleven types of ground-combat and transport unit: light, medium, and heavy armor, infantry, infantry FAC, infantry SAM, towed artillery, infantry HQ, armored HQ, truck, and light truck. We provide many more. Does this change outcomes? No. Effectively, all ground units are basically a protection class, defense strength, and sighting range, and the new units map to the existing ones. Does this change the experience? I think it does; the wider range of units enhances the gaming experience.

3. Infantry Weapons Platoon.

An infantry weapons platoon is not strictly necessary, but it adds to the look-and-feel, and we might consider giving them longer range in ground-to-ground combat. Having a single type is probably more appropriate than having separate types for grenade launchers, recoilless rifles, ATGMs, and machine guns. One might think that the .50 cal or DShK infantry AAA unit could serve as a machine-gun weapons platoon, but I would imagine most machine-gun weapons platoons have low tripods for ground combat rather than high ones for AAA.

4. Tanks.

Heavy Tanks.

- The WRG 1950–1985 rules give class IX or X (i.e., heavy) protection to the: Chieftain, Conqueror, EPC (Leclerc), IS-3, Leopard 2, Merkava 1, Shir 2 (Challenger 1), Stridsvagn 103, T-10, T-64, T-72, and T-80.
- Al-Kalid/VT-1A and Type 99: These are based on the Type 90-II prototype, which in turn is based on the T-72 (heavy).
- Ariete: This weighs 54 tonnes, only slightly less than the M1A1 (heavy) which also has a crew of four, and therefore presumably has similar protection.
- Arjun: The weights 59–68 tonnes and has crew of four, similar to the M1A1 (heavy), and so presumably has similar protection.
- Challenger 2: This is derived from the Challenger 1 (heavy).
- IS-2: This has a similar weight to the IS-3 (heavy) and a similar armament (122 mm D-25), so presumably similar protection.
- K1: This is derived from the M1 (heavy).
- K2: This presumably has at least the same protection as the K1 (heavy).
- M-84, PT-91, and T-90: These are derived from the T-72 (heavy).
- Merkava 2/3/4: These are derived from the Merkava 1 (heavy).
- T-14: This presumably has at least the same protection as the T-90 (heavy).
- T-84: This is derived from the T-80 (heavy).
- Type 10: This weighs 44 tonnes in its standard configuration and has a crew of three, and so presumably has protection at least equal to earlier tanks with similar weights such as the T-72 (heavy).
- Type 90: The has a crew of three and weights 50 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).
- Type 99: The has a crew of three and weights 51–55 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).

Medium Tanks.

- The WRG 1950–1985 rules give class IV to IIX (i.e., medium) protection to the: AMX-30, ASU-85, Centurion, Comet, Cromwell, ISU-122, ISU-152, Kanonen JPz, Leopard 1, M103, M4 Sherman, M46, M47, M48, M48, M60, Panzer 68, SU-100, T-34, T-54, T-55, T-62, TAM, Type 61, Type 74, and Vijayanta.
- M26: The M46 (medium) is largely an M26 with an improved engine.
- M-50 and M-51 Sherman: These are based on the M4 (medium) and have improved armament and engines.
- Panzer 61: This is essentially an early version of the Panzer 68 (medium).
- Type 59/69/79: The Type 59 was Chinese version of the T-54A (medium). The Type 69 and 79 had improved equipment, some of which was derived from captured T-62s, but the same level of protection.
- Type 85/88/96: These have new hulls and turrets. It’s not clear whether their level of protection is significantly improved over the Type 59/69/79 (medium).

- Type 15: Wikipedia gives little information on the armor, but the weight is 36 tonnes (with the armor package), and that seems to correspond more to a medium tank.
- Vickers MBT: This is similar to the Vijayanta (medium).

Light Tanks.

- The WRG 1950–1985 rules give class II and III (i.e., light) protection to the: AMX 10RC, AMX 13, Commando, Cougar, Fox, SpPz Luchs, M8, M24, M41, M114A2, M551 Panhard AML, Panhard EBR, PT-76, Saracen, Scimitar, and Scorpion
- Stryker MGS: With additional ceramic armor, this has protection against 14.5 mm but not heavier rounds, which I believe counts as III.
- Strv 74: This has a similar weight to the M41 (light).
- Type 62: Wikipedia give the hull armor as being similar to the M41 (light).
- Type 63: Wikipedia give the hull armor as being only 10 mm, which corresponds to light.
- Type 05: Based on Type 05 IFV (light).
- Type 08: Based on Type 08 IFV (light).

Open Tanks. Open tanks are uncommon, but we need an open target class for M3 half-track APCs, so it is easy to apply it here too.

- The WRG 1950–1985 rules give class I (i.e., open) protection to the: ASU-57 and SU-76.

5. Anti-Tank Missile Vehicles.

Anti-tank units are mainly chrome. However, in ground combat, we might give them more effectiveness against vehicular or towed targets and less against infantry targets.

Medium Anti-Tank Missile Vehicles.

- The WRG 1950–1985 rules give class IV (i.e., medium) protection to the: Raketen JPz.
- Jaguar: This is Raketen JPz (medium) upgraded with HOT missiles.

Light Anti-Tank Missile Vehicles.

- The WRG 1950–1985 rules give class II or III (i.e., light) protection to the: BRDM and Striker.
- Fennek MRAT: Based on the Fennek (light).
- FV438: This is derived from the FV432 APC (light).
- M150 and M901: These are derived from the M113 APC (light).
- M1134 Stryker ATGM: These are derived from the Stryker APC (light).
- VAB HOT: This is derived from the VAB APC (light).
- YP-408 PRAT: This is derived from the YPR-408 APC (light).
- YPR-765 PRAT: This is derived from the YPR-765 APC (light).

6. Reconnaissance Vehicles.

Again, reconnaissance vehicles are mainly chrome.

Light Reconnaissance Vehicles.

- The WRG 1950–1985 rules give class II or III (i.e., light) protection to the: BRDM-2, Ferret, Lynx (M113½ Lynx), and M114.
- LAV II Coyote: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.
- Fennek: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.

Light Reconnaissance Vehicles.

- M20: Open version of the M8 (light) with no turret and armed only with a .50-cal machine gun.

7. IFV Units.

We might distinguish APCs and IFVs in ground-to-ground combat. Bradley IFVs made mincemeat of Iraqi tank regiments; M113 APCs would not have had the same impact.

Medium IFVs.

- The WRG 1950–1985 rules give class IV (i.e., medium) protection to the: Marder.
- BMP-3: Wikipedia says the BMP-3 has frontal protection from 30 mm rounds at 200 m.
- Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
- Strf 9040/CV90: Wikipedia states that the CV90 has frontal protection against 30 mm APFSDS rounds. This is considerably more than typical light AFVs.

– LAV III (add-on-armor): Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.

– M2A2/M2A3/M2A4: Wikipedia states their these M2A2 also has protection against 30 mm APDS rounds and RPGs. The M2A3 and M2A4 have visually similar armor.

– Puma: Wikipedia says the frontal arc has protection against medium-caliber rounds and shaped-charge projectiles. With additional armor, the sides gain similar protection.

Light IFVs.

– The WRG 1950–1985 rules give class II or III (i.e., light) protection to the: AMX-10P, BMD, BMP-1, Commando, Dutch AIFV (YPR-765), FV432 (FV432 RARDEN), HS.30 (SPz Lang), M113 (M113 FSC), Spz Kurz, UK MICV (Warrior), and XM723 (M2/M3 Bradley).

– BMP-2: Wikipedia says the BMP-2 has similar protection to the BMP-1.

– LAV-25: Wikipedia says this only has protection against small-arms rounds.

– LAV III: Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.

– Pbv 301: Wikipedia states the armor ranges from 8 to 50 mm, with upper front armor being 34 mm (20 mm at 55 degree). This seems to be light.

– Piranha IFV: Presumably similar protection to LAVs.

– Stryker ICVD: This is derived from the Stryker (light).

– Type 86: Wikipedia states this is reverse-engineered from the BMP-1.

– Type 92: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 100 meters and 25 mm rounds at 1000 meters. This seems to correspond to light armor. The weight is also 21 tonnes, which is considerable less than most medium IFVs.

– Type 05: Wikipedia states that it has protection against small-arms rounds and some protection against 12.7 mm rounds. This corresponds to light armor.

– Type 08: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 200 meter. This corresponds to light armor.

– VCBI: Wikipedia says this only has protection against 14.5 mm rounds.

There are many variants of the Piranha, and the in-service date is a guess.

8. APC Units.

Heavy APCs. These are converted tanks.

– Achzarit: Wikipedia states that this based on the T-54/T-55 (medium), but has additional composite armor. Its weight at 44 tonnes is much heavier than the T-55 at 36 tonnes, despite the lack of turret, which suggests the additional armor is quite significant. Therefore, we consider it to be heavy.

– Namer: This is based on the Merkava IV (heavy).

– Nakpadon: This is an improved version of the Nagmashot and Nagmachon. Wikipedia states that its weight is around 55 tonnes, which strongly suggests its armor qualifies as heavy.

– Puma: The photos in Wikipedia show additional armor. Its weight at 50 tonnes is almost identical to the Centurion, despite the lack of turret, which suggests the additional armor is quite significant.

The in-service dates for the Nakpadon (1998) and Puma (1993) are guesses.

Medium APCs.

– Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.

– Nagmashot: Based on the Centurion (medium).

– Nagmachon: Based on the Centurion (medium).

The in-service dates for the Nagmashot (1990) and Nagmachon (2000) are guesses.

Light APCs.

– The WRG 1950–1985 rules give class II or III (i.e., light) protection to the: BTR-50PK, BTR-60PB, BTR-152K, Commando, Grizzly, FV432, LVTP-7, M113, M59, Saracen, Spartan, VAB, and YPR-408.

– Bronco/Warthog: Wikipedia says it has protection against 7.62 mm rounds.

– BVS10/Viking: Wikipedia says it has protection against small-arms fire and can have additional armor to protect against 14.5 mm rounds.

– BTR-70: Wikipedia states the frontal armor is only 9 mm.

– BTR-80: Wikipedia states that this (and previous models) only have protection from small-arms fire.

– LAV II Bison: Wikipedia states that the Coyote is protected against small-arms rounds, and the Bison seems to be similar.

– LVTP-5: Wikipedia states that the maximum armor is 16 mm.

– M75: Wikipedia states the maximum armor is 38 mm. This is more than the thickness of some light tanks, but it is almost vertical.

– OT-64 SKOT: Wikipedia states the maximum armor is 13 mm.

– OT-810: Wikipedia states the maximum armor is 14.5 mm.

– Piranha APC: Presumably similar protection to LAVs.

– M1126 Stryker: Wikipedia states it has some protection against 14.5 mm rounds from the front and all-round protection against small-arms fire.

– Saxon: Wikipedia states that it is protected against small-arms rounds.

– TPz Fuchs: Wikipedia states that it is protected against small-arms rounds.

– Type 63: Wikipedia states that it is protected against small-arms rounds.

– Type 77: Wikipedia states that it is based on the PT-76 (light), and its weight of 15-5 tonnes matches other light APCs.

– Type 89: Wikipedia states that it is protected against small-arms rounds.

– Type 08: Based on Type 08 IFV (light).

There are many variants of the Piranha, and the in-service date is a guess.

Open APCs.

– The WRG 1950–1985 rules give class I (i.e., open) protection to the: AT-P, BTR-40, BTR-50P, BTR-60P, BTR-152A, and Half-tracks (M3).

– SKP/VKP/SKPF/VKPF: Wikipedia states the armor is 8 to 20 mm, which is light.

9. Armored Mortars.

Regarding the protection class:

– FV432: based on FV432 (light).

– M106/M125/M1064: based on M113 (light).

– M1129 Stryker MCV: based on Stryker (light).

We might consider these as open.

10. Armored Artillery.

Light Armored Artillery.

– The WRG 1950–1985 rules give class II (i.e., light) protection to the: Abbot, AMC-155GCT (AMX-30 AuF1), L33, M52, M108, M109, SP.73 (2S3), SP.74 (2S1), and VK.155 (Bkan 1).

– 2S19 Msta-S: Wikipedia states this all-round protection from 15 mm of steel armor. This corresponds to light armor.

– AS-90: Wikipedia states this has up to 17 mm of steel armor. This corresponds to light armor.

– Panzerhaubitze 2000: Wikipedia states this all-round protection from 14.5 mm of steel armor. This corresponds to light armor.

Open Armored Artillery. Regarding the protection class:

– The WRG 1950–1985 rules give class I (i.e., open) protection to the: AMX-105A, M7, and M44.

– The M-50 has frontal and side armor, like the M7 and unlike for example the M107 and M108.

– The M40 also has frontal and side armor. The rear door is probably armored too, but is open when the gun is used.

11. Armored Rocket Artillery.

Light Armored Rocket Artillery.

– M270: I can find no definite information on the armor protection.

– BM-1: According to Wikipedia, the T-72 hull was chosen to give adequate protection close to the front lines. However, it is not clear whether the rocket launcher has adequate protection.

12. Tracked Carriers.

All the listed tracked carriers are stated as being unarmored by Wikipedia.

13. Tracked Artillery.

– The WRG 1950–1985 rules give class I (i.e., open) protection to the: AMX.155F3 (AMX-13 F3), M107, and M110. However, the only crew member protected seems to be the driver. The other crew members travel and operate the gun without protection.

– Similar considerations apply to the: 2S4, 2S7 (although some of the crew travel under armor), M40, M41, and Mk F3. In all of these, the gun mount is open.

14. Mobile Artillery.

Regarding the protection class:

– CAESAR: Some models have lightly-armored cabins, but the truck as a whole does not seem to be protected.

– Archer: The crew and engine compartments is protected against small-arms rounds, but the gun does not seem to be protected. Unlike the CAESAR, the gun is operated by the crew from the protected cabin. This is close to reaching the level of protection of light armored artillery.

15. Mobile Rocket Artillery.

Regarding the protection class:

– M142 HIMARS: The cabin has some protection, but the rest of the truck seems to be unprotected.

– BM-30: The cabin appears to have some protection, but the rest of the truck seems to be unprotected.

– Earlier Soviet vehicles appear to be completely unarmored.

16. Mobile Rocket Artillery.

Regarding the protection class:

– M752 Lance: This seems to be based on the unarmored M548.

– AMX-30 Pluton: The vehicle is based on an AMX-30 hull and seems to be armored, but the missile is exposed.

Tracked missile artillery is mainly included to serve as a target.

17. Mobile Missile Artillery Batteries.

Regarding the protection class, none of these have complete protection against small arms.

Missile artillery is mainly included to serve as a target.

18. HQ Units.

I wondered if HQ platoons were also used at the company level, but looking at the *Air Strike* scenarios they are included at a ratio of about one HQ platoons to nine normal platoons, so they seem to be battalion HQs.

19. British Army Units.

The generic British Army formations from 1960 and 1985 are from the document “Modern British TOEs” by Richard A. Rinaldi (2002).

20. Soviet Units.

These are from US Army FM 100-2-3 from 1991.

21. Rules.

Much of this section should be included in the 3A rules, but it’s here for the moment.

22. Squads and Sections as Damaged Platoons.

So for example sections/squads have 2D/1D damage resilience and +1/+2 modifiers to AAA attacks. Similar modifiers should be applied to ground-to-ground attacks.

23. Damage Resilience in *Air Strike*.

There is a form of damage resilience in *Air Strike*, in that units whose counters have white text are only able to take 1D of damage before being killed (see the last paragraph of rule 28). Such units include infantry SAMs, armored CCUs, and many vehicle SAMs. TSOH also mentions units being able to take only 2D damage, but it doesn’t have any relevant units since it attaches SAM teams to infantry platoons rather than having them as separate (see below) and doesn’t have a separate FAC unit. In both cases, the idea seems to have been incompletely implemented in the rules. I’m generalizing and homogenizing the idea. This is important as we need sections for mobile AAA units with two vehicles and for the Oerlikon GDF (as Skyguard can only control two guns and not a full battery). To a large degree, a section is equivalent to a platoon that has taken 1D and a squad is equivalent to a platoon that has taken 2D, so the mechanics of the game are really not impacted.

24. Hard Targets.

Light, medium, and heavy armor correspond to defense strengths of 4, 6, and 8 in *Air Strike*.

Light armor give protection from 14.5 mm and smaller round. Medium armor gives protection from autocannons and guns up to about 90 mm. Heavy armor gives protection against guns of 100 mm or more. Obviously, this is a gross simplification as each vehicle has its own strengths and weaknesses and its armor varies between its front, side, rear, and top. Nevertheless, this rough classification seems to map to the vehicles depicted in scenarios in *Air Strike* and *Eagles of the Gulf*.

The protection level of tanks and other AFVs are based on the armor classes in the Wargames Research Group 1950–1985 rules: open is class I, light is class II to III, medium is class IV to IIX, and heavy is class IX and X. More modern tanks are guesses, and I’m willing to be corrected.

25. Open Targets.

The BTR-50P APC and the M3 half-tracked APC are examples of open targets. In the 1956 and 1967 Arab-Israeli Wars where both sides used M3 half-tracks. If an M3 is attacked by an air-burst or napalm bomb, the crew and passengers will suffer more or less as if they were in the open.

26. Soft Targets.

Examples of soft targets include infantry, towed artillery, unarmored vehicles, and armored vehicles that do not fully protect their personnel and/or weapons such as the M-107 and M-110 tracked artillery vehicles.

Nothing like the M-107 or M-110 is included in *Air Strike*, but again I would assert that it is tracked mainly for mobility. Effectively, these are towed artillery installed in an exposed position on a tracked hull and have almost no protection for most of the crew.

Most tracked SAM vehicles are classified as hard targets in *Air Strike* and *Eagles of the Gulf*, despite many of them having completely exposed missiles. If an air-burst bomb explodes above a SAM-4 launcher, it is not going to shrug it off. I think many SAM vehicles are tracked mainly for *mobility* and perhaps for crew protection, but in terms of a mission-kill on their weapon systems, they are soft targets. I’ve not changed anything yet, but we need to think about this.

27. Representation.

The representation is based on NATO symbols, including those for infantry, armor, anti-armor, air defense, guns, missiles, mortar, tube artillery, and rocket artillery. One addition is that a rectangle denotes an unarmored vehicle such as a truck or tracked carrier. A truck or mobile unit has two wheel symbols in the lower position, whereas a tracked carrier has a tracked symbol. The mortar symbol is modified to be simply an open circle, for compactness.

All vehicular units have either an armored or unarmored vehicle symbol, all infantry units have an infantry symbol, and towed units have none of these.

The letter T in the central position indicates a transport capacity (e.g., APCs, IFVs, cargo trucks, and tracked cargo carriers). Words and letters in the lower position indicate variants on a basic type: the letters H, M, L, and O on an armored unit indicate heavy, medium, light, and open armored protection; the word WPN or FAC on an infantry unit indicates a weapons or FAC unit; and the letter H on an infantry SAM unit indicates a heavy version of the launcher (e.g., the LML version of Starstreak).

28. Mobility.

The inspiration for this rule was this photograph of the Mitla Pass from 1967, which shows trucks destroyed on and just off a road. Also, in the the Falklands War roads could be traversed by Landrovers and AMLs, but off-road terrain was impassable to everything except the CVR(T)s and Bv 202s despite being open.

29. Mobility Restrictions.

When trucks have good mobility, the rule is: don’t put trucks or towed weapons in forests unless there is a road. When trucks have poor mobility, the rule is effectively: if the terrain is difficult, keep your trucks and towed weapons near roads even in clear terrain.

30. Truck Mobility.

For example, a scenario set in the open desert of southern Iraq might specify that trucks have good mobility, but one set in the Falkland Islands might specify that they have poor mobility. Similarly, a scenario might specify that cargo trucks have poor mobility, but mobile artillery units have good mobility. Finally, a scenario might not assign an explicit transport unit to a towed artillery battery, but might state that it has been transported by helicopter and so can be placed as if it had a transport unit with good mobility.

31. Modifying Mobility Classes.

One common modification is to explicitly indicate the hexes occupied by each ground unit; in this case this rule is largely superfluous (although if we allow ground units to move, then the restrictions can still be relevant).

Panhard AMLs are wheeled armored vehicles and so normally have unrestricted mobility. Nevertheless, in the Falklands War they were restricted to roads and trails, and this corresponds to poor mobility. A scenario could fix this problem by explicitly giving them poor mobility.

32. Composite Platoons.

This rule comes from TSOH. It doesn’t have counters for infantry SAMs, so in scenarios V-11 and V-22 it says treat infantry platoons as if they have an

infantry SAM squad. This rule helps with stacking (since attached squads do not count for stacking) and with keeping your precious FACs and SAM teams anonymous. It also works nicely with the transport rule, since attached squads do not need their own transport but can ride with the platoon.

33. **Damage to Attached Squads.**

The roll of 4– seems about right. The probability of killing an attached squad is 40% for 1D and 64% for 2D. Those numbers are fairly good approximations to 1/3 and 2/3. If the roll was 3–, they would be 30% and 51%, and that seems a little low.

Are the damage rolls secret?

34. **Transportation.**

The idea of transporting units and cargo comes from the “Thunder and Fire” scenario from *Eagles of the Gulf*. The rules here simply generalize it slightly, incorporate the idea of mobility, and add rules for mounting and dismounting.

35. **Mobility.**

Being placed according to the mobility of its transport allows a tracked transport to place a towed unit in a forest hex.

36. **Mounting and Dismounting Infantry.**

The rules for mounting and dismounting are completely new. I’m very open to suggestions about this.